

**Problem 1.**

*Solution:*

- (a) Pseudo-code for generating  $N_t$

$$N_t = \min \left\{ n : \sum_{i=1}^n U_i > t \right\}, \quad U_i \sim \text{Unif}(0, 1),$$

where  $t = 1$  gives  $N_1$  and  $t = 2$  gives  $N_2$ .

```

1 Algorithm Generate_Nt(t):
2   sum = 0
3   n = 0
4   while sum <= t:
5     sample U ~ Uniform(0,1)
6     sum = sum + U
7     n = n + 1
8   end while
9   return N_t = n

```

- (b) Point estimate and 95% Confidence Interval To estimate  $\mathbb{E}[N_t]$ , generate  $R$  independent copies of  $N_t$ , say

$$N_t^{(1)}, N_t^{(2)}, \dots, N_t^{(R)}, \quad R = 100, 1000, 10000.$$

**Point estimate:**

$$\bar{N}_t = \frac{1}{R} \sum_{r=1}^R N_t^{(r)}.$$

**Sample standard deviation:**

$$S_t = \sqrt{\frac{1}{R-1} \sum_{r=1}^R \left( N_t^{(r)} - \bar{N}_t \right)^2}.$$

**95% Confidence interval for  $\mathbb{E}[N_t]$ :**

$$\left[ \bar{N}_t - z_{0.975} \frac{S_t}{\sqrt{R}}, \bar{N}_t + z_{0.975} \frac{S_t}{\sqrt{R}} \right],$$

where  $z_{0.975} = 1.96$ . That is,

$$\boxed{95\% \text{ CI: } \left[ \bar{N}_t - 1.96 \frac{S_t}{\sqrt{R}}, \bar{N}_t + 1.96 \frac{S_t}{\sqrt{R}} \right].}$$

- (c) Explanation:

By the *Law of Large Numbers*, the sample mean  $\bar{N}_t$  is a consistent estimator of  $\mathbb{E}[N_t]$ .

By the *Central Limit Theorem*, when  $R$  is large,  $\bar{N}_t$  is approximately normal, which justifies the 95% confidence interval:

$$\bar{N}_t \pm 1.96 \frac{S_t}{\sqrt{R}}.$$

Since  $N_2 \geq N_1$  almost surely, we expect  $\mathbb{E}[N_2] > \mathbb{E}[N_1]$ , which is confirmed by the results.

(d) Results:

|   |                                                    |
|---|----------------------------------------------------|
| 1 | t=1, R=100: mean=2.7500, 95% CI=(2.5940, 2.9060)   |
| 2 | t=1, R=1000: mean=2.6770, 95% CI=(2.6246, 2.7294)  |
| 3 | t=1, R=10000: mean=2.7066, 95% CI=(2.6897, 2.7235) |
| 4 | t=2, R=100: mean=4.8700, 95% CI=(4.6245, 5.1155)   |
| 5 | t=2, R=1000: mean=4.6430, 95% CI=(4.5644, 4.7216)  |
| 6 | t=2, R=10000: mean=4.6588, 95% CI=(4.6346, 4.6830) |

## Problem 2.

*Solution:*

$$\text{Var}(X) = \sigma^2 = 2 \quad \Rightarrow \quad \sigma = \sqrt{2}.$$

We have confidence level  $1 - \alpha = 0.95$ , hence

$$1 - \frac{\alpha}{2} = 0.975 \quad \Rightarrow \quad z_{0.975} = 1.96.$$

By the *Central Limit Theorem (CLT)*, for large  $n$ ,

$$\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_j$$

is approximately normal with mean  $\mu$  and variance  $\frac{\sigma^2}{n}$ . Therefore, a 95% confidence interval for  $\alpha$  is

$$\bar{X}_n \pm z_{0.975} \frac{\sigma}{\sqrt{n}}.$$

We require the half-width of the confidence interval to be at most 0.05:

$$z_{0.975} \frac{\sigma}{\sqrt{n}} \leq 0.05.$$

Substituting  $\sigma = \sqrt{2}$  and  $z_{0.975} = 1.96$  gives

$$n \geq \left( \frac{z_{0.975} \sigma}{0.05} \right)^2 = \left( \frac{1.96 \times \sqrt{2}}{0.05} \right)^2 = 3073.28.$$

$$\boxed{n = 3074}$$

**Problem 3.**

*Solution:*

- (a) Find a 95% confidence interval for  $\theta$ .

$$n = 30$$

$$\bar{X} = \frac{1}{30} \sum_{i=1}^{30} X_i \approx 122.1667$$

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^{30} (X_i - \bar{X})^2} \approx 18.2758$$

$$z_{0.975} = 1.96$$

$$\frac{S}{\sqrt{n}} = \frac{18.2758}{\sqrt{30}} \approx 3.338$$

$$\text{Margin of error: } z_{0.975} \frac{S}{\sqrt{n}} = 1.96 \times 3.338 \approx 6.539$$

Hence, the 95% confidence interval is

$$\bar{X} \pm z_{0.975} \frac{S}{\sqrt{n}} = [122.1667 - 6.539, 122.1667 + 6.539] = [115.6267, 128.7066].$$

- (b) How many additional replications are required for 99% confidence and accuracy  $\pm 0.5$ ?  
Given:

$$\varepsilon = 0.5, \quad z_{0.995} = 2.5758.$$

We require

$$n \geq \left( \frac{z_{0.995} S}{\varepsilon} \right)^2 = \left( \frac{2.5758 \times 18.2758}{0.5} \right)^2 \approx 8865.$$

Hence, the smallest integer  $n = 8865$ . Since we already have 30 replications, the number of *additional* replications needed is

$$8865 - 30 = 8835.$$

**Problem 4.**

*Solution:*

$$(X - 1)^2 < 30 \quad \Rightarrow \quad |X - 1| < \sqrt{30} \approx 5.477 \quad \Rightarrow \quad X \in (-4.477, 6.477).$$

$$n = 40$$

|       |    |   |    |   |   |   |    |
|-------|----|---|----|---|---|---|----|
| $X_i$ | 2  | 4 | 5  | 6 | 8 | 9 | 24 |
| Times | 11 | 5 | 10 | 2 | 7 | 4 | 1  |

Values  $X_i = 2, 4, 5, 6$  fall within  $(-4.477, 6.477)$  and are therefore counted as *successes* (Bernoulli trials).

**Point estimate:**

$$\hat{p} = \frac{11 + 5 + 10 + 2}{40} = \frac{28}{40} = 0.7.$$

**95% Confidence Interval:**

$$\hat{p} \pm z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Substitute  $z_{0.975} = 1.96$ ,  $\hat{p} = 0.7$ , and  $n = 40$ :

$$\hat{p} \pm 1.96 \sqrt{\frac{0.7(0.3)}{40}} = 0.7 \pm 1.96 \times 0.0724 = 0.7 \pm 0.1418.$$

95% CI: [0.558, 0.842].

### Problem 5.

*Solution:*

(a)

$$n = 20, \quad \alpha = 0.01, \quad 1 - \frac{\alpha}{2} = 0.995, \quad z_{0.995} = 2.5758.$$

Let  $Y = X^2$ . Then  $\mathbb{E}[X^2] = \mathbb{E}[Y]$ .

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{20} \sum_{i=1}^{20} X_i^2 \approx 181.85.$$

$$S_Y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2} \approx 180.0314.$$

We require

$$N \geq \left( \frac{z_{0.995} S_Y}{\varepsilon} \right)^2 = \left( \frac{2.5758 \times 180.0314}{2} \right)^2 \approx 53761.43.$$

Hence,

Smallest integer total:  $N = 53762$ .

Additional replications required:  $53762 - 20 = 53742$ .

(b) Assumptions:

- The simulation outputs are i.i.d. replications of  $X$ .
- $Y = X^2$  has a finite variance.
- By the Central Limit Theorem, the sample mean of  $Y$  is approximately normal when  $n$  is large (rule of thumb:  $n \geq 30$ ).
- The point estimate of variance from the initial 20 runs is used as a plug-in for  $\sigma^2$ .
- The sample-size formula is valid only when the obtained  $n$  is sufficiently large.
- “Margin of error” is interpreted as the half-width of the confidence interval.